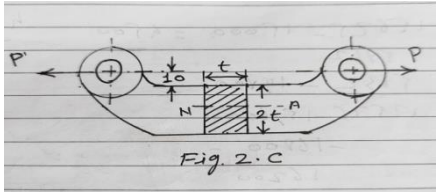


DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE Supplementary Summer Examination-2023 Course: B. Tech. Branch : Mechanical Engineering Semester : V Subject Code & Name: BTMEC 503 Machine Design –I Max Marks: 60 Date: 11/8/23 Duration: 3 Hr.			
Instructions to the Students: 1. All the questions are compulsory. 2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question. 3. Use of non-programmable scientific calculators is allowed. 4. Assume suitable data wherever necessary and mention it clearly.			
		(Level/CO)	Marks
Q. 1	Solve Any Two of the following.		12
A)	Write answers of following MCQ'S 1) The following standard(s) is (are) used in Mechanical engineering design. a) Standards for materials, (b) Standard for testing of products (c) Standards for Fits, Tolerances and surface finish of component (d) All of the above. 2) Aesthetic considerations in designs are related to--- a) Colour combinations, b) Curved surfaces, c) Easy to handle d) External appearance of the product 3) A Splined shaft is one having---- a) Single Key B) Multiple Keys C) Keys as a integral part of shaft 4) For cast iron components -----is considered to be the failure criterion. a) Ultimate tensile strength , b) Yield Strength, c) Endurance limit d) None of the mentioned 5) A beam subjected to bending moment undergoes the following stresses. a) Compressive , b) Tensile, c) Both compressive & tensile d) None of the mentioned 6) Determine the width of the cotter used in cotter joint connecting two rods subjected to axial load of 50kN and permissible shear stress in cotter is 50N/(mm ²). Given thickness of cotter=10mm a) 50 mm , b) 100 mm, c) 150 mm d) 25mm	1/1	6
B)	Explain the term stress concentration and the methods to reduce the same.	1/2	6
C)	Explain the aesthetic and ergonomic consideration in design & its importance .	1/2	6
Q.2	Solve Any Two of the following.		12
A)	Explain the construction and working principle of Knuckle joint with the help of neat diagram.	2/2	6
B)	A cotter joint is to be used to connect two rods made of plain carbon steel 30 C8 having $S_{yt} = 400 \text{ N/mm}^2$ & subjected to a tensile load of 125 KN. The cotter is made from steel plate with thickness of 15 mm. Calculate the dimensions of socket and cotter assuming following data:- The yield strength in compression is twice the yield strength in tension. The yield strength in shear is 50% of the yield strength in tension. The factor of safety is 5.	2/3	6
C)	An offset link subjected to a force P of 30 KN is as shown in the fig. 2 c. it made of cast iron FG300 having $S_{ut} = 300 \text{ N/mm}^2$ and the F.S is 3 . Determine the dimensions of cross section of the link.	2/4	6
			

Q. 3	Solve Any Two of the following.		12
A)	Explain the following terms:- 1) Notch Sensitivity 2) Endurance limit and factors affecting it.	2/2	(2X3)=6
B)	A rotating bar made of steel 40 C8 having $S_{ut}=600 \text{ N/mm}^2$ is subjected to completely reversed bending stress. The corrected endurance limit of the bar is 320 N/mm^2 . Calculate the fatigue strength of the bar for a life of 90,000 cycles.	2/5	6
C)	A spherical pressure vessel having inner diameter of 500 mm is welded from steel plate. The welded joints are sufficiently strong. The plate is made from cold drawn steel 20C8 ($S_{ut} = 400 \text{ N/mm}^2$ and $S_{yt} = 245 \text{ N/mm}^2$). The vessel is subjected to internal pressure varying from zero to 6 N/mm^2 . The expected reliability is 50 % and F.S. is 3. The size factor is 0.85. the vessel is expected to withstand infinite no of stress cycles. Calculate the thickness of the plate. Use the following data:- $K_a = 0.82$, $K_b = 0.85$, & $K_c = 1.0$	3/3	6
Q.4	Solve Any Two of the following.		12
A)	what is the function of key? Explain its types with the help of neat diagram.	2/2	6
B)	A shaft 50 mm in diameter is used to transmit 30 KW power at 250 rpm to the gear mounted on it. A square key is used to connect the gear on shaft. The key is made of steel 50 C4 having $S_{yt} = 450 \text{ N/mm}^2$ and the F.S is 3. For the key material yield strength in compression can be assumed to be equal to the yield strength in tension. Determine the dimensions of the key.	2/6	6
C)	A rigid flange coupling is used to transmit 25 KW power at 750 rpm. There are 4 bolts at pitch circle diameter of 125 mm. The shaft and bolts are made of 40 C 8 ($S_{yt} = 400 \text{ N/mm}^2$) and factor of safety is 3. Assume $S_{sy} = 0.577 S_{yt}$ and determine the diameter of shaft and bolts.	3/6	6
Q. 5	Solve Any Two of the following.		12
A)	Explain the following terminology used in springs with the help of neat diagrams :- 1) Free length 2) compressed length 3) solid length	2/3	(3X2) 6
B)	A tension spring balance consists of helical tension spring which is attached to fixed support at one end and carries masses at free end. The pointer attached to free end moves on a scale and indicates mass. The length of the scale is 100 mm which is divided into 50 equal divisions. Each division on scale indicates 0.5 kg . The max. Capacity of the balance is 50 kg. and the spring index is 6 . The spring is made of oil hardened and tempered steel wire of grade SW ($G = 81370 \text{ N/mm}^2$ & $S_{ut} = 1600 \text{ N/mm}^2$) . The permissible shear stress in the spring wire is 50 % of the ultimate tensile strength. Design the spring and give its specification.	2/5	6
C)	The lead screw of a lathe machine is provided with Acme threads having 50 mm outside diameter and 8 mm pitch. The axial force exerted by lead screw is 2500 N. The collar subjected to thrust in the carriage has 110 mm outside diameter and 55 mm inside diameter and the lead screw rotates at 30 rpm. The coefficient of friction for screw is 0.15 and that for collar is 0.12 . Determine 1) The power required to drive the screw and 2) The efficiency of the lead screw.	2/4	6
	*** End ***		